

DAHD-MSLM application: Lorenz96

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1. Numerical setup

To demonstrate Multi-layer Stuart-Landau oscillator stochastic modeling (MSLM) capabilities enabled by DAHD for Lorenz96 (L96) model data (Chekroun and Kondrashov, 2017), we utilize here more accurate and computationally-efficient DAHD formulation in frequency domain (Kondrashov et al., 2020, 2026) rather than its original time domain formulation (Chekroun and Kondrashov, 2017). Here DAHD is realized numerically by an eigendecomposition of the Hermitian cross-spectral density matrix (Kondrashov et al., 2020, 2026). The resulting eigenvectors yield an orthonormal set of spatiotemporal data adaptive harmonic modes (DAHMs), associated with a specific temporal frequency, which in turn allows to rank these modes by their captured energy and across frequencies, and allow accurate space-time reconstruction.

The Lorenz 96(L96) model is a periodic system of coupled nonlinear ordinary differential equations given by:

$$\frac{du_i}{dt} = u_{i-1}(u_{i+1} - u_{i-2}) - u_i + F_i, i = 0, \dots, J - 1 \quad (1)$$

where $J = 40$ and F_i is specified value of the deterministic forcing. This model has become popular as a toy model of the atmosphere to test various methods in the weather and climate science.

When the forcing is sufficiently strong (in this study $F_i = 6$), the L96 becomes a prototype of turbulent dynamical system which exhibits features of weakly chaotic dynamics. L96 model in Eq. (1) is integrated by a 4th order Runge-Kutta numerical method with a time step $dt = 0.01$. Excluding transitional behavior and subsampling every 10th time step, $N = 11900$ points is retained for further analysis. The L96 solution is dominated by highly irregular spatio-temporal traveling wave pattern, as shown in top panel of Fig.1.

The embedding window M is the main parameter of DAHD. The choice of M is based on two competing goals: (i) to obtain reliable estimates of autocorrelations from noisy and short datasets; and (ii) to resolve the dataset's frequency domain for identification purposes with sufficient accuracy. M should be larger than typical decorrelation times in the data; it is the slowest temporal scale characterized by DAHD. We chose here $M = 160$ which results in a total number $dM = 6400$ of DAHD eigenvalues.

Figure 2 shows the DAHD spectrum composed of the eigenvalues (red full circles) and obtained for the L96 dataset, showing pronounced low-frequency peak corresponding to the main period of traveling wave. Each of the DAHMs $\mathbf{E}_j$, i.e. $1 \leq j \leq dM'$ (where $M' = 2M - 1$) represents a data-adaptive spatio-temporal pattern associated with a fixed temporal frequency; the latter are equally spaced at intervals of $1/(M' - 1)$ in the Nyquist interval $[0, 0.5]$. Moreover, each temporal

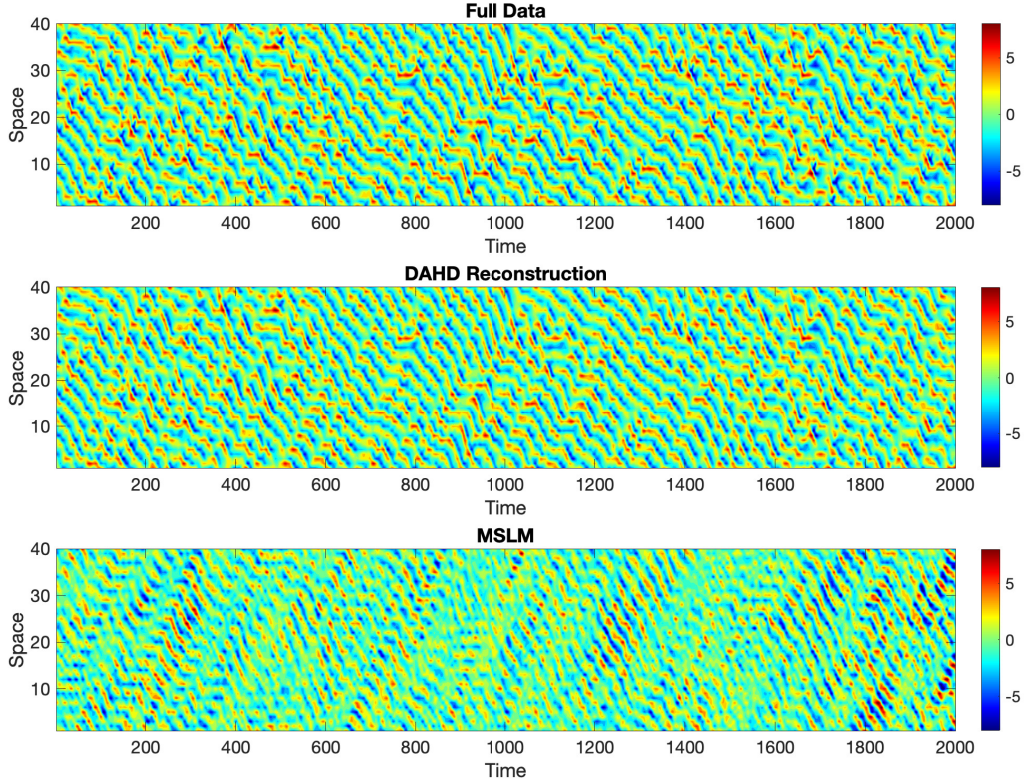


Figure 1: top panel - L96 reference data; middle panel - DAHD reconstruction obtained by using leading 20 DAHD eigenelements in the $f=[0, 0.1]$ (blue colored dots in Fig. 2); (bottom panel) MSLM stochastic simulation.

frequency f is associated with d DAHM pairs and eigenvalues, except at $f = 0$, where there is only one DAHM per eigenvalue.

The time-embedded spatial structure of the DAHMs associated with a top spectral DAHD eigenvalue at selected frequencies (black dots in Fig. 2), are shown in Fig. 3. This structure conveys information about the amplitude modulation across spatial channels, and the figure demonstrates that indeed the DAHMs for each pair, except at zero frequency, are in phase quadrature, i.e. shifted by one quarter of the associated period. Projection of original spatio-temporal L96 data onto DAHMs yields a set of narrow-band time series pairs (roughly in phase quadrature) $(p_k(t), q_k(t))$, which are referred to as data-adaptive harmonic coefficients (DAHCs) shown also in Fig. 3.

Figure 1 (middle panel) shows the space-time patterns of the harmonic reconstruction components (HRCs) summed across frequencies by using DAHMs and DAHCs associated with leading 20 eigenelements at 30 frequencies in $[0, 0.1]$ range (blue dots in Fig. 2), i.e. $20 \times 30 = 600$ that represents just a fraction of total DAHD spectral eigenelements – $d \cdot M = 6400$. As we can see these patterns match quite well in frequency and phase those of the reference coherent components.

The original d -dimensional spatiotemporal field $\mathbf{x}(t)$ with its multiple time scales decomposed into DAHCs is then modeled by a network frequency-ranked multilayer nonlinear stochastic Stuart–Landau oscillators (MSLM) (Kondrashov et al., 2018; Chekroun and Kondrashov, 2017):

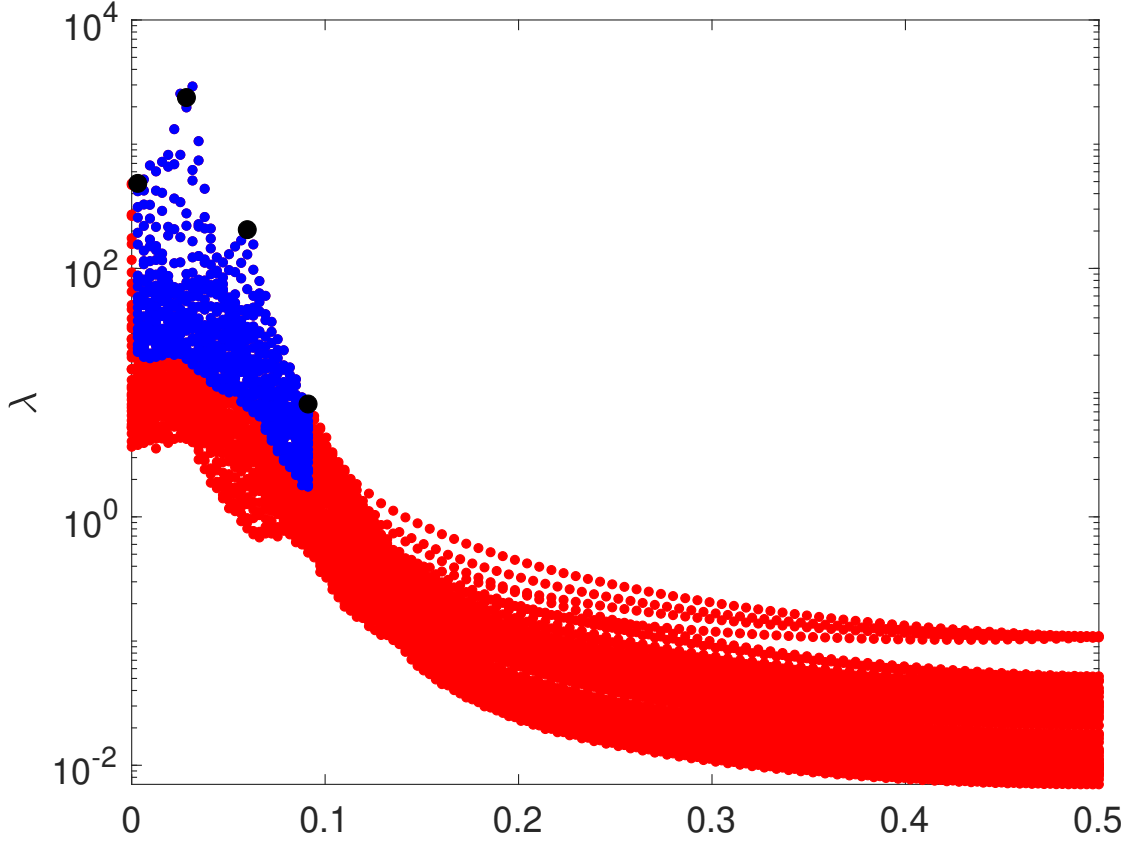


Figure 2: DAHD spectrum of the L96 dataset with “bartlett” weighting applied. Each red full circle corresponds to a DAHD eigenvalue with pair of data-adaptive DAHD modes (DAHMs) that represent the same temporal frequency f but are time-shifted so as to be in phase quadrature. The frequencies of the DAHMs are equally spaced between 0 and 0.5, and the total number of DAHM pairs in each frequency bin is equal to the number of channels $d = 40$ in the dataset (except zero frequency where there is only one DAHM per eigenvalue). Blue - top 20 eigenelements used for the MSLM simulation.

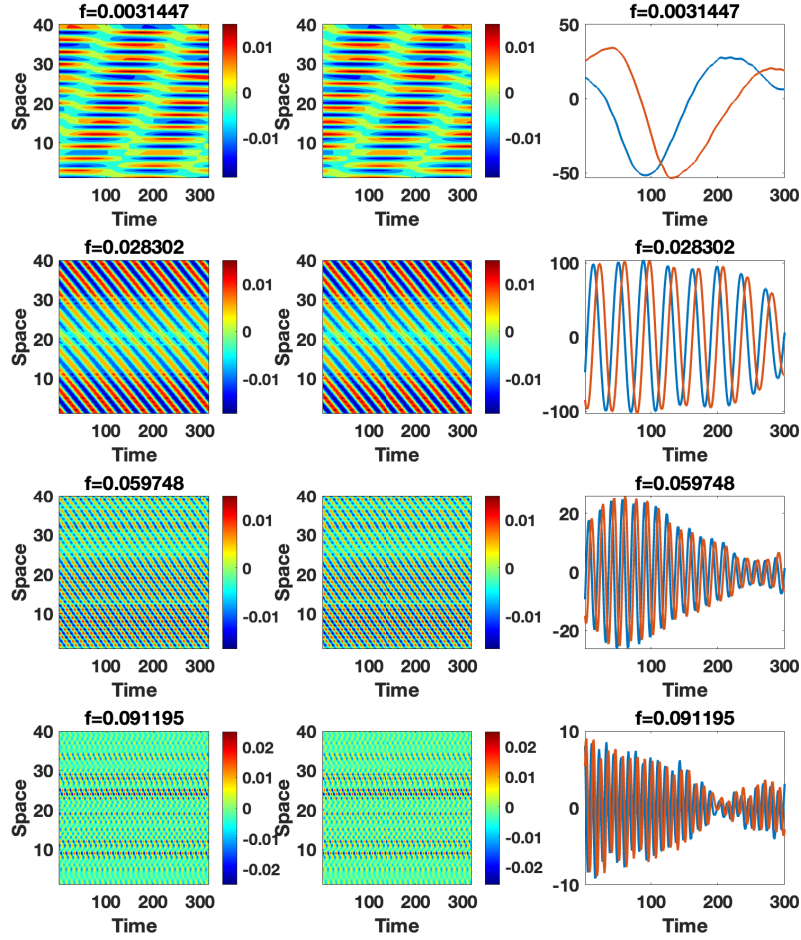


Figure 3: DAHMs (let and middle panel) and corresponding data-adaptive harmonic coefficients (DAHCs) (right panel) for the leading eigenvalues at the four frequencies (black dots in Fig. 2).

$$\begin{aligned}
dp_k/dt &= \beta_k(f)p_k - \alpha_k(f)q_k - \sigma_k(f)p_k(p_k^2 + q_k^2) + \sum_{i \neq k}^d a_{ik}(f)p_i + \sum_{i \neq k}^d b_{ik}(f)q_i + \epsilon_k^p, \\
dq_k/dt &= \alpha_k(f)p_k + \beta_k(f)q_k - \sigma_k(f)q_k(p_k^2 + q_k^2) + \sum_{i \neq k}^d c_{ik}(f)p_i + \sum_{i \neq k}^d d_{ik}(f)q_i + \epsilon_k^q,
\end{aligned} \tag{2}$$

where $1 \leq k \leq d$, model parameters are estimated by pairwise multiple linear regression with linear constraints on $\alpha_k(f)$ and $\beta_k(f)$ to ensure antisymmetry for the linear coupling within the given pair, as well as equal and positive values $\sigma_k(f) > 0$ to ensure numerical stability while $(\epsilon_k^p, \epsilon_k^q)$ is a pairwise correlated noise estimated from regression residuals and modeled by additional linear layers of the models (see (Kondrashov et al., 2018; Chekroun and Kondrashov, 2017) for details), and at the last level is approximated by white noise.

Simulated MSLM field in physical space is obtained after convolving corresponding DAHMs with the simulated DAHCs, here obtained by using five levels of the MSLM (including the main level of Eq.(2). The resulting field is shown in Fig. 1 (bottom panel) and exhibits qualitatively common features with the original field (middle and top panels). Fig. 4 compares autocorrelation function (ACF) (in time) of each channel as well as probability density function (PDF), followed by an averaging over the channels. Overall, the time-variability of the L96 flow is reproduced reasonably well by MSLM.

2. Matlab routines

runmsmDAHD4_L96 – main script of this example.

MSLM_INT.m– general purpose routine for fitting MSLM model and long-term integration

statglob.m – auxillary routine to compare summary statistics

Requires also DAHD routines from **DAHD4example**

3. Best practices/FAQ

- Embedding window M should be larger than typical decorrelation times in the data; it is the slowest temporal scale characterized by DAHD. For the very long time series such as in this example we use $M \ll N$.
- The Hermitian DAHD formulation (Kondrashov et al., 2020, 2026) closely resembles a SPOD/SEOF approach (Towne et al., 2018), with the key difference being that the latter considers Welch’s overlapped averaging periodogram estimate of the cross-spectral density (CSD) matrix, while DAHD is based on a correlogram estimate; both in turn can be understood as approximating dynamic mode decomposition (DMD) in ensemble context and relating to Koopman theory (Tu et al., 2014).
- Applying weighting to time-lagged cross-correlations prior to computing CSD matrix reduces bias at large lags, helps to make it positive-definite, and ensures its eigenvalues are positive, also helps with numerical stability of MSLM.

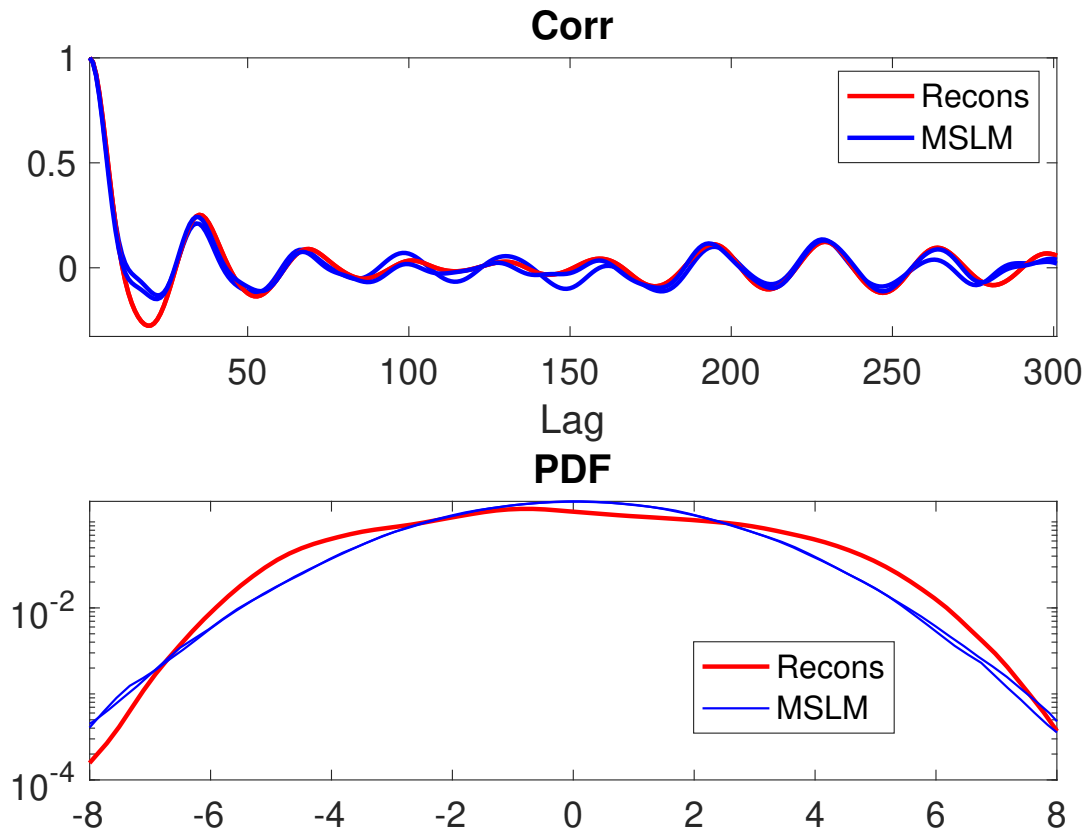


Figure 4: Summary statistics: - autocorrelations (top panel) and PDF (bottom panel) for original L96 data (red) and MSLM stochastic realizations (blue) showing reasonably good match.

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